Question 1: What happens if we change the number of components?

Change PCA(n_components=2) to PCA(n_components=3).

- How many features are present after reduction?
- Does the number of samples change?
- Does the total variance retained increase or decrease?

Question 2: What happens if standardization is removed?

Replace X_scaled with X in the PCA transformation.

- Does the explained variance ratio change?
- Why might features with larger numerical scales influence PCA more strongly?
- Is standardization always necessary?

Question 3: What happens if we retain more components?

Try 1, 2, 3, 5 and 10 principal components.

- How does the cumulative explained variance change?
- How many components are needed to retain at least 90% of the variance?
- Is reducing to two components always the best choice?

Question 4: What happens to the visualization?

Plot the first two principal components and observe the three wine classes.

- Are the classes clearly separated?
- What information might be lost when reducing 13 features to 2?
- Does PCA use the class labels to create the components?

Question 5: Does PCA select existing features?

- Are the two principal components original Wine dataset columns?
- What is the difference between feature selection and feature extraction?
- Why is PCA considered a feature extraction technique?